

Domain-Specific Data Analytics & Data Science Notes

(Banking & Financial Services)

01. Overview — Banking & Financial Services (BFS)

What is this domain?

The Banking & Financial Services (BFS) domain covers institutions that manage money, credit, risk, and capital for individuals and businesses. This includes:

- **Retail Banking** – savings/current accounts, personal loans, credit cards, mortgages
- **Corporate/Commercial Banking** – business loans, trade finance, cash management
- **Investment Banking** – capital raising, M&A advisory, underwriting
- **Wealth Management & Private Banking** – portfolio management, advisory services
- **NBFCs (Non-Banking Financial Companies)** – lending without full banking licenses
- **Fintech** – digital-first lending, payments, neobanks, BNPL (Buy Now Pay Later)
- **Capital Markets** – trading, brokerage, asset management

Why data analytics/science matters here

Banking is fundamentally a **data-driven business** — every decision (who to lend to, how much interest to charge, which transaction is fraudulent, which customer will churn) is backed by data. Key reasons data roles are critical:

1. **Risk is core to the business model** — banks profit by pricing risk correctly (credit risk, market risk, operational risk).
2. **Heavy regulation** — data must be accurate, auditable, and reported to regulators (RBI, SEC, FCA, Basel Committee, etc.).
3. **High transaction volume** — millions of transactions daily generate massive datasets suited for analytics/ML.
4. **Fraud & security** — real-time detection of anomalies is a matter of financial survival.
5. **Customer-centric growth** — cross-sell, up-sell, and retention are driven by behavioral analytics.

Industry context & key regulatory bodies

Region Regulator(s)

India RBI (Reserve Bank of India), SEBI, IRDAI

USA Federal Reserve, OCC, FDIC, SEC, CFPB

UK/EU FCA, PRA, ECB, EBA

Global Basel Committee on Banking Supervision (BCBS)

Important frameworks/regulations a BFS data professional should be aware of:

- **Basel II/III/IV** – capital adequacy, risk-weighted assets (RWA)
- **AML (Anti-Money Laundering) & KYC (Know Your Customer)**
- **GDPR / Data Privacy laws** – for customer data handling
- **IFRS 9 / CECL** – provisioning for expected credit losses
- **PCI-DSS** – payment card data security

Typical business functions a Data Analyst/Scientist supports

- Credit Risk & Underwriting
- Fraud Detection & Prevention
- Collections & Recovery
- Customer Analytics (acquisition, retention, cross-sell)
- Regulatory Reporting & Compliance
- Marketing & Campaign Analytics
- Treasury & Asset-Liability Management (ALM)
- Anti-Money Laundering (AML) Monitoring

Why this domain is attractive for Data Analysts/Scientists

- High demand for analytical talent due to regulatory pressure and digital transformation
- Rich, structured, and semi-structured datasets (transactions, applications, bureau data, call logs)
- Direct link between models and revenue/loss — high visibility of analytical work

- Strong career path: Analyst → Senior Analyst → Data Scientist → Risk Manager/Quant → Chief Risk/Data Officer
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02. Role of a Data Analyst/Scientist in Banking & Financial Services

This is the core file — it walks through **what a Data Analyst/Data Scientist actually does**, step-by-step, across the **entire project lifecycle** in a banking/financial services setting.

A. Day-to-Day Responsibilities (Snapshot)

- Pulling and validating transactional/customer data from core banking systems
 - Building and monitoring dashboards for risk, fraud, or portfolio health
 - Writing SQL queries for ad-hoc business requests from Risk, Credit, or Compliance teams
 - Building/maintaining ML models (credit scoring, fraud detection, churn prediction)
 - Preparing regulatory reports (NPA reports, AML alerts, Basel disclosures)
 - Presenting insights to business stakeholders (Credit Heads, Risk Committees)
 - Conducting root-cause analysis when KPIs move unexpectedly (e.g., spike in defaults)
 - Collaborating with Data Engineering to ensure data pipeline quality
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B. End-to-End Project Lifecycle in Banking Analytics

Below is a **detailed, step-by-step breakdown** of how a typical analytics/data science project is executed in a bank — using a **Credit Risk / Loan Default Prediction** project as a running example, since it's one of the most common BFS use cases.

Step 1: Business Problem Understanding

- Meet with stakeholders (Credit Risk team, Collections team, Product Owner) to understand the pain point.
 - Example: *"Our default rate on personal loans has increased by 3% in the last two quarters. We need to identify high-risk applicants before disbursal."*
- Translate the business problem into an analytical problem statement:
 - Business ask → *Reduce defaults*
 - Analytical framing → *Build a binary classification model to predict probability of default (PD) at the time of loan application*
- Define success metrics with stakeholders (e.g., reduce default rate by X%, improve approval accuracy, maintain approval volume)

Step 2: Data Collection & Understanding

- Identify data sources:
 - **Internal:** Core Banking System (CBS), Loan Origination System (LOS), transaction history, past repayment behavior
 - **External:** Credit bureau data (CIBIL/Experian/Equifax), KYC documents, income proof, employment data
- Understand data dictionaries, table relationships (ER diagrams), and data ownership
- Check data availability across time periods (need enough history of "good" vs "bad" loans)
- Define the **target variable** clearly:
 - Example: Bad = 1 if a customer has missed 90+ days of payment (DPD90+) within 12 months of disbursal; else Good = 0

Step 3: Data Extraction

- Write SQL queries / use tools (Hive, Spark SQL, Snowflake) to extract required data from data warehouse/lake
- Pull data at the right **observation point** (avoid data leakage — only use info available *at the time of application*, not after)
- Join multiple sources: application data + bureau data + past internal behavior data

- Sample data appropriately (stratified sampling if class imbalance is high, e.g., few defaulters vs many good customers)

Step 4: Data Cleaning & Preprocessing

- Handle missing values:
 - Bureau data often has missing fields for "thin file" customers (no credit history) — treat separately
- Handle outliers (e.g., unrealistic income values, negative loan tenure)
- Standardize formats (dates, currency, categorical labels across systems)
- Deduplicate customer records (same customer across multiple loan applications)
- Handle class imbalance (typically default rate is 2–10%) using:
 - Oversampling (SMOTE), undersampling, or class-weighting techniques

Step 5: Exploratory Data Analysis (EDA)

- Univariate analysis: distribution of income, loan amount, age, credit score
- Bivariate analysis: default rate by income bracket, employment type, loan tenure, geography
- Correlation analysis between bureau attributes and default flag
- Identify seasonality/trends (e.g., higher defaults post-festive season loans)
- Visualize using bar charts, box plots, WOE (Weight of Evidence) bins — very common in credit risk

Step 6: Feature Engineering

- Create domain-specific derived variables, e.g.:
 - **Debt-to-Income Ratio (DTI)**
 - **Credit Utilization Ratio**
 - **Number of active loans**
 - **Days since last delinquency**
 - **Vintage of oldest credit line**
 - **Bureau score bands**

- Perform **WOE (Weight of Evidence) & IV (Information Value)** transformation — a banking-standard technique to convert variables into a scorecard-friendly format
- Select variables based on Information Value (IV), correlation, and business judgment (avoid using proxy-discriminatory variables like religion/gender to remain compliant with fair-lending laws)

Step 7: Model Building

- Split data into train/test/out-of-time (OOT) samples — OOT validation is critical in banking to test model stability over time
- Common algorithms used:
 - **Logistic Regression** – still the industry standard for credit scorecards (interpretable, regulator-friendly)
 - **Decision Trees / Random Forest / XGBoost / LightGBM** – for higher accuracy, often used internally alongside scorecards
 - **Survival Analysis** – for time-to-default modeling
- Build a **credit scorecard** by converting model coefficients into points (using PDO — Points to Double the Odds methodology)
- Tune hyperparameters using cross-validation

Step 8: Model Validation

- Evaluate using banking-relevant metrics:
 - **KS Statistic (Kolmogorov-Smirnov)** – measures separation between good/bad populations
 - **AUC-ROC**
 - **Gini Coefficient**
 - **Population Stability Index (PSI)** – checks if scoring population has shifted over time
 - **Confusion Matrix, Precision/Recall** at chosen cutoff
- Validate against **Out-of-Time (OOT)** sample to ensure model generalizes to future data, not just historical patterns

- Perform **stress testing** — how does the model perform in adverse economic scenarios (recession, rate hikes)?

Step 9: Model Governance, Documentation & Regulatory Review

- Document model methodology, assumptions, and limitations (Model Risk Management requirement)
- Submit for internal **Model Validation team** review (independent of the build team — common in regulated banks)
- Ensure compliance with fair lending regulations (no disparate impact on protected classes)
- Get sign-off from Risk Committee before deployment

Step 10: Deployment

- Convert model logic into production code (Python/SQL/PMML/rules engine)
- Integrate model into the Loan Origination System (LOS) so scores are generated in real-time during application processing
- Set up A/B testing or a "shadow mode" period — run new model in parallel with existing rules without impacting decisions, to compare performance

Step 11: Monitoring & Maintenance

- Set up ongoing monitoring dashboards:
 - Score distribution drift (PSI)
 - Default rate by score band (does the model still separate risk correctly?)
 - Approval rate trends
- Schedule periodic model **recalibration or redevelopment** (typically annually, or sooner if performance degrades)
- Track model performance against a **model risk tiering framework** (high/medium/low risk models get different monitoring frequency)

Step 12: Reporting & Stakeholder Communication

- Present results to Credit Risk Committee, senior leadership using clear visuals (score distribution, expected loss reduction, approval rate impact)
- Prepare regulator-facing documentation if required (e.g., under Basel IRB approach)

- Continuously gather feedback from Collections/Credit teams on real-world model performance
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C. This Lifecycle Applies (with variations) to Other BFS Use Cases

The same structured approach (Problem → Data → EDA → Feature Engineering → Modeling → Validation → Deployment → Monitoring) applies to:

- **Fraud Detection** – real-time transaction scoring instead of application-time scoring
 - **Customer Churn Prediction** – for retail banking/credit card attrition
 - **AML Transaction Monitoring** – rule-based + ML hybrid systems flagging suspicious activity
 - **Collections Prioritization** – predicting which delinquent customers to contact first
 - **Marketing Propensity Models** – predicting likelihood of loan/product uptake
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D. Tools Used Across the Lifecycle (Quick Reference)

Stage	Common Tools
Data Extraction	SQL, Hive, Spark, Snowflake
EDA & Feature Engineering	Python (Pandas, NumPy), R, Excel
Modeling	Python (Scikit-learn, XGBoost, Statsmodels), SAS
Visualization/Reporting	Power BI, Tableau, Excel
Model Deployment	Python/Flask APIs, PMML, rules engines
Monitoring	Custom dashboards, Airflow-scheduled reports

03. Key Metrics & KPIs — Banking & Financial Services

A Data Analyst/Scientist in BFS must be fluent in the following categories of metrics. These are used constantly in reporting, model evaluation, and business reviews.

1. Credit Risk & Lending Metrics

Metric	Meaning
NPA (Non-Performing Asset) Ratio	% of loans where borrower has stopped repaying (90+ days past due)
DPD (Days Past Due)	Number of days a payment is overdue (e.g., DPD30, DPD60, DPD90)
Probability of Default (PD)	Likelihood a borrower defaults within a given time horizon
Loss Given Default (LGD)	% of exposure the bank loses if a default occurs
Exposure at Default (EAD)	Total value the bank is exposed to at the time of default
Expected Credit Loss (ECL)	$PD \times LGD \times EAD$ — used under IFRS 9/CECL provisioning
Approval Rate	% of loan applications approved
Credit Score / Bureau Score	Standardized score (e.g., CIBIL 300–900) representing creditworthiness
Vintage Analysis	Default trends tracked by the month/quarter a loan was originated
Roll Rate	% of accounts moving from one delinquency bucket to a worse one (e.g., DPD30 → DPD60)

2. Fraud & AML Metrics

Metric	Meaning
Fraud Detection Rate	% of actual fraud cases correctly flagged
False Positive Rate	% of legitimate transactions incorrectly flagged as fraud
Alert-to-Case Conversion Rate	% of AML alerts that turn into confirmed suspicious activity reports (SARs)

Metric	Meaning
Chargeback Rate	% of transactions disputed/reversed by customers

3. Customer & Retail Banking Metrics

Metric	Meaning
Customer Acquisition Cost (CAC)	Cost to acquire a new customer
Customer Lifetime Value (CLV)	Total projected revenue from a customer over their relationship with the bank
Churn/Attrition Rate	% of customers who close accounts or stop using services
Cross-sell Ratio	Average number of products held per customer
Net Promoter Score (NPS)	Customer satisfaction/loyalty measure
Active-to-Dormant Ratio	% of accounts that are actively transacting vs inactive

4. Profitability & Financial Metrics

Metric	Meaning
Net Interest Margin (NIM)	Difference between interest earned and interest paid, relative to assets
Cost-to-Income Ratio	Operating costs as a % of income generated
Return on Assets (ROA) / Return on Equity (ROE)	Profitability relative to assets/equity
CASA Ratio	Current Account Savings Account ratio — cheap source of funds for banks

5. Model Performance Metrics (Statistical)

Metric	Meaning
KS Statistic	Max separation between cumulative good vs bad distributions
Gini Coefficient / AUC-ROC	Overall discriminatory power of a model

Metric	Meaning
Population Stability Index (PSI)	Measures how much the scored population has shifted from model development to current
Characteristic Stability Index (CSI)	Similar to PSI, but for individual variables
Precision, Recall, F1-score	Standard classification metrics, especially for fraud models

6. Regulatory & Capital Metrics

Metric	Meaning
Capital Adequacy Ratio (CAR)	Bank's capital relative to its risk-weighted assets (Basel requirement)
Risk-Weighted Assets (RWA)	Assets weighted by risk level, used for capital calculations
Liquidity Coverage Ratio (LCR)	Ability to cover short-term obligations with liquid assets

04. Common Datasets & Data Sources — Banking & Financial Services

Internal Data Sources

Source System	Typical Data
Core Banking System (CBS)	Account balances, transaction history, account opening/closing dates
Loan Origination System (LOS)	Loan application details, applicant demographics, approval/rejection status

Source System	Typical Data
Loan Management System (LMS)	Repayment schedules, EMI history, delinquency status
Card Management System	Credit/debit card transactions, limits, utilization
CRM Systems	Customer interactions, service requests, complaints
Collections System	Contact history, promise-to-pay records, recovery status
Internal Fraud/AML Systems	Flagged transactions, case investigation outcomes

External Data Sources

Source	Typical Data
Credit Bureaus (CIBIL, Experian, Equifax, CRIF)	Credit score, active loans, repayment history, credit inquiries
KYC/Identity Verification Providers	Aadhaar/PAN verification (India), identity documents
Income Verification Services	Bank statement analysis, Form 16/salary slips, GST data (for SMEs)
Macroeconomic Data	Interest rates, inflation, unemployment rate (used in stress testing)
Alternative Data (Fintech/NBFC use)	Utility bill payments, mobile/telecom usage, e-commerce transaction data

Typical Fields/Attributes an Analyst Works With

Customer/Application Data:

- Age, gender, income, employment type, employer name, residence type, tenure at current job/residence

Loan/Account Data:

- Loan amount, tenure, interest rate, EMI, disbursal date, product type, collateral details

Bureau Data:

- Credit score, number of active trade lines, total outstanding debt, DPD history, number of hard inquiries in last 6 months

Transaction Data:

- Transaction amount, merchant category code (MCC), transaction type, channel (ATM/POS/online), timestamp, location

Behavioral/Repayment Data:

- Payment history (on-time/late), bounce/NACH failure history, partial payments

Data Characteristics Specific to Banking

- **Highly sensitive/PII data** — requires masking, encryption, and strict access controls
 - **Regulatory retention requirements** — data often must be stored for 7–10 years
 - **Imbalanced target classes** — defaults/fraud are rare events (often <5% of population)
 - **Time-series nature** — most banking data has a temporal dimension (vintage, seasonality)
 - **Data leakage risk** — must carefully separate "information known at decision time" vs "information known later"
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05. Tools & Technologies — Banking & Financial Services

Programming Languages

- **SQL** — non-negotiable; used daily for data extraction from data warehouses
- **Python** — Pandas, NumPy, Scikit-learn, XGBoost, Statsmodels for modeling and analysis
- **R** — still used in some risk/actuarial teams for statistical modeling

- **SAS** – widely used in large, traditional banks for credit scoring and regulatory reporting (legacy but still very relevant)

Data Storage & Processing

- **Data Warehouses:** Snowflake, Teradata, Google BigQuery, Amazon Redshift
- **Big Data:** Hadoop, Apache Spark/PySpark (for processing large transaction datasets)
- **Data Lakes:** Azure Data Lake, AWS S3

Visualization & BI Tools

- **Tableau** – common for risk/portfolio dashboards
- **Power BI** – widely used in retail banking reporting
- **Excel** – still heavily used for ad-hoc analysis, pivot tables, VBA macros
- **QlikView/Qlik Sense** – used in some legacy banking environments

Modeling & Statistical Tools

- **Scikit-learn, XGBoost, LightGBM** – for ML-based risk/fraud models
- **Statsmodels** – for logistic regression, statistical testing
- **SAS Enterprise Miner** – for credit scorecard development in traditional banks
- **MATLAB** – occasionally used in quantitative/market risk teams

Specialized Banking/Risk Software

- **SAS Fraud Management / FICO Falcon** – fraud detection platforms
- **Moody's Analytics / SAS Risk Management** – credit risk & regulatory capital tools
- **Actimize / NICE Actimize** – AML transaction monitoring
- **Bloomberg Terminal / Reuters Eikon** – market data (investment banking/treasury roles)

Workflow & Orchestration

- **Apache Airflow** – scheduling data pipelines/reports
- **Git/GitHub, GitLab** – version control for code and models
- **Jira/Confluence** – project tracking and documentation

Cloud Platforms

- **AWS, Azure, GCP** – increasingly used for scalable data processing and ML deployment, though many traditional banks still rely on on-premise infrastructure due to regulatory/data residency requirements

Model Deployment & MLOps

- **Flask/FastAPI** – wrapping models as APIs for integration into loan origination systems
 - **Docker/Kubernetes** – containerizing models for deployment
 - **MLflow** – model versioning and experiment tracking (increasingly adopted)
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06. Real-World Use Cases — Banking & Financial Services

1. Credit Risk & Underwriting

- **Credit Scoring/Scorecard Development** – predicting probability of default for loan applicants
- **Loan Approval Automation** – rules + ML-based auto-decisioning for small-ticket loans
- **Limit Setting** – determining credit card/loan limits based on risk profile
- **Portfolio Risk Segmentation** – bucketing customers into risk tiers for differentiated treatment

2. Fraud Detection & Prevention

- **Real-time Transaction Fraud Scoring** – flagging suspicious card/UPI transactions instantly
- **Account Takeover Detection** – identifying unusual login/access patterns
- **Merchant Fraud Detection** – identifying fraudulent merchant behavior in payment networks
- **Synthetic Identity Fraud Detection** – identifying fake identities created using real+fake data combinations

3. Anti-Money Laundering (AML)

- **Transaction Monitoring Systems** – flagging structuring, layering, or unusual fund movement patterns
- **Sanctions Screening** – matching customers/transactions against global watchlists
- **Suspicious Activity Report (SAR) Prioritization** – ranking AML alerts by risk to optimize investigator time

4. Collections & Recovery

- **Collections Prioritization Models** – predicting which delinquent customers are most likely to pay if contacted
- **Best Time/Channel to Contact** – optimizing collection call timing and channel (SMS/call/email)
- **Settlement Amount Optimization** – predicting the minimum settlement a customer would accept

5. Customer Analytics & Marketing

- **Customer Segmentation** – grouping customers by behavior/value for targeted offers
- **Next-Best-Offer/Product Recommendation** – predicting which product a customer is likely to buy next (credit card, personal loan, insurance)
- **Churn Prediction** – identifying customers likely to close accounts or switch banks
- **Campaign Response Modeling** – predicting likelihood of response to marketing campaigns

6. Regulatory & Compliance Reporting

- **Basel/IFRS 9 Provisioning Calculations** – computing expected credit losses for financial reporting
- **Stress Testing** – simulating portfolio performance under adverse economic scenarios
- **Regulatory Dashboard Automation** – automating recurring compliance reports (NPA reports, liquidity reports)

7. Treasury & Asset-Liability Management

- **Interest Rate Risk Modeling** – analyzing impact of rate changes on bank's balance sheet
- **Liquidity Forecasting** – predicting cash flow needs to maintain regulatory liquidity ratios

8. Investment Banking & Capital Markets (for analysts in this vertical)

- **Algorithmic Trading Signal Generation**
- **Portfolio Risk Analytics (VaR — Value at Risk)**
- **M&A Due Diligence Data Analysis**

Example Interview-Style Problem Statements

- *"Build a model to predict which customers are likely to default on their credit card payments in the next 3 months."*
 - *"Design a real-time fraud detection system for UPI transactions that minimizes false positives."*
 - *"How would you segment our retail banking customers to identify cross-sell opportunities for personal loans?"*
 - *"Our collections team wants to know which delinquent customers to call first — how would you build this prioritization model?"*
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07. ML & Statistical Techniques — Banking & Financial Services

Classification Models (Most Common)

- **Logistic Regression** – industry standard for credit scorecards due to interpretability and regulatory acceptance
- **Decision Trees / Random Forest** – used for both risk and fraud modeling, good for capturing non-linear relationships
- **Gradient Boosting (XGBoost, LightGBM, CatBoost)** – widely used for fraud detection and higher-accuracy risk models
- **Support Vector Machines (SVM)** – occasionally used for fraud classification

Techniques Specific to Credit Risk

- **WOE (Weight of Evidence) & IV (Information Value)** – standard technique to bin and transform variables for scorecard models
- **Scorecard Development (PDO methodology)** – converting logistic regression outputs into point-based credit scores
- **Survival Analysis (Cox Proportional Hazards)** – modeling time-to-default rather than binary default
- **Markov Chain Models** – used in roll-rate analysis (modeling transitions between delinquency states)

Fraud & Anomaly Detection Techniques

- **Isolation Forest, One-Class SVM, Autoencoders** – unsupervised anomaly detection for fraud where labeled fraud data is limited
- **Graph-based Models / Network Analysis** – detecting fraud rings by analyzing relationships between accounts/entities
- **Rule-based Engines + ML Hybrid Systems** – combining business rules with ML scores (very common in production fraud systems)

Time-Series & Forecasting

- **ARIMA/SARIMA, Exponential Smoothing** – for forecasting cash flow, liquidity needs, deposit trends
- **Prophet** – used for business-friendly forecasting of transaction volumes

Customer Analytics Techniques

- **Clustering (K-Means, Hierarchical Clustering)** – for customer segmentation
- **RFM Analysis (Recency, Frequency, Monetary)** – customer value segmentation
- **Uplift Modeling** – measuring incremental impact of marketing campaigns/interventions
- **Collaborative Filtering** – for product recommendation engines

AML-Specific Techniques

- **Rule-based Scenario Detection** – structuring, layering pattern rules
- **Network/Graph Analytics** – identifying complex money laundering networks across accounts
- **NLP** – for screening adverse media/news related to customers (negative news screening)

Statistical Testing & Validation

- **KS Test, Gini/AUC** – model discriminatory power
- **Population Stability Index (PSI) / Characteristic Stability Index (CSI)** – monitoring model/data drift
- **Chi-Square Test, T-tests** – feature significance testing
- **A/B Testing** – for testing new risk policies or marketing strategies before full rollout

Model Governance Concepts (Unique to Banking)

- **Model Risk Management (MRM)** – frameworks like SR 11-7 (US Fed guidance) for model validation
 - **Explainability (SHAP, LIME)** – increasingly required even for ML models to satisfy regulators on "right to explanation"
 - **Fair Lending Analysis** – statistical testing to ensure models don't discriminate against protected classes (disparate impact analysis)
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08. Resources — Banking & Financial Services

Public Datasets for Practice

- **Lending Club Loan Data (Kaggle)** – real-world peer-to-peer lending data for credit risk modeling
- **German Credit Data (UCI Repository)** – classic small dataset for credit scoring practice
- **Give Me Some Credit (Kaggle)** – credit risk/default prediction competition dataset
- **Credit Card Fraud Detection Dataset (Kaggle, ULB)** – anonymized real transaction data for fraud modeling
- **Home Credit Default Risk (Kaggle)** – large-scale alternative credit data competition

- **IEEE-CIS Fraud Detection (Kaggle)** – large fraud detection dataset with rich features
- **Santander Customer Transaction Prediction (Kaggle)** – anonymized banking data for classification practice

Recommended Courses

- Credit Risk Modeling in Python (DataCamp)
- Financial Risk Management Specialization (Coursera – NYU)
- Basel III & Risk Management courses (Coursera/edX)
- FRM (Financial Risk Manager) certification material – for deep risk domain knowledge
- Fraud Analytics using Descriptive, Predictive, and Social Network Techniques (Coursera – KU Leuven)

Books

- *"Credit Risk Scorecards"* – Naeem Siddiqi (industry standard reference for scorecard building)
- *"The Basel Handbook"* – for regulatory capital concepts
- *"Fraud Analytics Using Descriptive, Predictive, and Social Network Techniques"* – Bart Baesens

Communities & Blogs

- Kaggle competition discussion forums (search "credit risk", "fraud detection")
- Towards Data Science (Medium) – search banking/fintech analytics articles
- Risk.net – industry news on risk management and regulation
- LinkedIn groups: "Credit Risk Analytics", "Banking Analytics Professionals"

Regulatory Reference Material

- Basel Committee on Banking Supervision (BCBS) publications – bis.org
- RBI Master Circulars (for India-specific banking regulations) – rbi.org.in
- IFRS 9 Financial Instruments standard summary documents

GitHub Repositories for Reference

- Search GitHub for: credit-scorecard, fraud-detection-model, loan-default-prediction for open-source implementation examples

This completes the Banking & Financial Services domain notes. Update this file as you discover new resources during your learning journey.